Start Here, Navyashree B H

Your mission: produce the real, measured computer-vision evidence for our research paper

Hi Navyashree B H! This folder is your complete guide to the **computer vision (CV) part** of our AgriVision paper. You will build the camera circuit, collect plant images, train two models, measure how well they work, and document everything so Shreyas can drop your numbers straight into the paper. Read the files in order. Nothing here needs you to be an expert: every step has a checklist, and file 05 has ready-made prompts for AI tools.

RULE: The paper is judged on technical depth and honesty. Every number, figure and sentence about CV must come from something you actually ran. If a result looks bad, record it anyway. A bad result that is honestly explained is worth more than a perfect-looking result nobody can reproduce.

1. What the paper needs from you

The final vision story is a **two-stage pipeline**: (1) a detector decides *is this a plant, and where?*; (2) a classifier looks at the close-up and decides *healthy or diseased (and which disease)?*. The camera sits on a **pan-tilt** mount so the rover can look around. We start with **tomato**.

#	Deliverable (evidence for the paper)	Goes into paper section	Folder / file
D1	Working ESP32-CAM + pan-tilt bench circuit, photos and test log	III-A, IV-A	02 hardware guide, camera_tests.csv
D2	Dataset: public tomato + your own field/camera images, with metadata	V (datasets)	03 dataset guide, dataset_log.csv
D3	Stage 1 plant detector: precision, recall, mAP50, mAP50-95, latency	VI, Table II	04 training guide, detector_results.csv
D4	Stage 2 classifier: accuracy and macro-F1 on 3 test sets, 3 seeds	VI, Table III	04 training guide, classifier_results.csv
D5	Cross-domain test (lab-trained model on field/ESP32-CAM images)	VI discussion	04 training guide
D6	Speed: camera snapshot time, model inference time	VI, Table IV	latency_bench.py
D7	Figures: circuit photo, confusion matrix, PR curve, example detections, failure cases	III, IV, VI	05 results guide
D8	Literature table: 15-20 verified papers	II (related work)	07 literature guide, literature_table.csv

2. The deadline plan (today is 1 Nov 2026; paper is due 14 Nov, 23:59 AoE)

The plan works backwards so that Shreyas has 4 days to write. **Hard handover: end of 10 Nov.** If something will not be ready, say so early; we then mark it 'in progress' in the paper instead of faking it.

Dates	Goal	Done when
1-2 Nov	Read guides 00-07. Set up Colab, GitHub/Drive folder. Flash the stock CameraWebServer example and see live video.	Live camera view on laptop; screenshot saved
3-4 Nov	Build the pan-tilt circuit; run servo + camera tests. Download public tomato data. Start literature table.	pan_tilt_tests.csv has 10+ rows; datasets downloaded
4-8 Nov	Collect your own images (field if possible, else potted plants) with the metadata sheet. Expert label check.	300+ images per class, plant-level IDs recorded
6-9 Nov	Train detector and classifier (3 seeds). Evaluate on in-domain, cross-domain and own-camera sets.	detector_results.csv and classifier_results.csv filled
9-10 Nov	Latency tests, figures, evidence folder, 1-page results summary. Handover to Shreyas.	Evidence checklist (file 06) fully ticked

Dates	Goal	Done when
11-13 Nov	Support paper writing: answer questions, regenerate figures, fix numbers.	Paper read-through by you
13 Nov eve	Submit early; do NOT wait for the last hour (portal can be slow).	Confirmation e-mail saved

3. Files in this folder

- **00_Start_Here** (this file): mission, deliverables, schedule.
- **01_Pipeline_and_Concepts**: the two-model idea explained simply, plus the metrics you will report.
- **02_Hardware_Circuit_Guide**: ESP32-CAM + pan-tilt wiring, power, flashing, tests, mounting.
- **03_Dataset_Collection_Guide**: public datasets, your field protocol, labelling, plant-level splits.
- **04_Model_Training_Guide**: step-by-step training and evaluation with the starter code.
- **05_Results_and_Documentation_Guide**: what to log, figure list, folder structure, naming.
- **06_Paper_Evidence_Checklist**: tick-list of everything Shreyas needs from you.
- **07_Literature_Search_Guide**: how to find and verify references.
- **08_AI_Prompt_Library**: copy-paste prompts for ChatGPT / Claude / Gemini / Copilot.
- **Starter Code**: capture, split, training, evaluation, latency scripts. **Templates**: CSV logs to fill.

4. Ground rules

1. **Never invent or round up a result.** Copy numbers from the script output.
2. **Keep raw outputs.** Save every training log, JSON and screenshot. Name them with date and run id.
3. **Split by plant, not by image** (explained in file 03). This is the single most important rule.
4. **Cite everything you use**: datasets, libraries, papers. Write the source down when you download it.
5. **Keep the paper anonymous.** Never put names, college or logos inside figures or screenshots meant for the paper.
6. **Ask early.** If blocked for more than 2 hours, message Shreyas with a screenshot and what you tried.